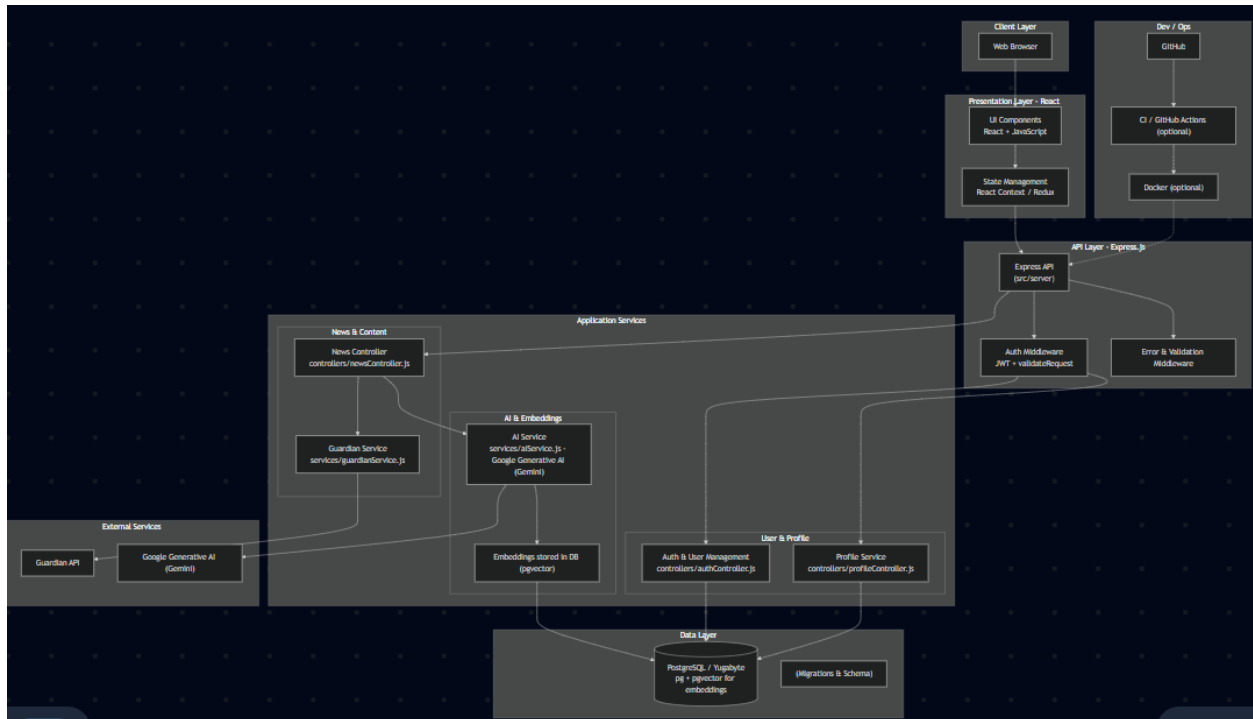


News 4 U High-Level Architecture V3

Architecture Diagram



Client Layer

- Consists solely of Web Browser
- Initially had plans for Screen Readers for more accessibility but wasn't able to implement successfully while trying to get the AI model integrated

Dev/Ops

- Working with Github for development and shared repositories
- Helps support continuous integration
- Docker containers used to create testing environments for development

Presentation Layer

- UI components through React/JS
- React has frameworks to help make state and context management easier

API Layer

- Using Express API for client/server connection
- Auth middleware to ensure client authorized for specific services
- Error middleware to ensure errors sent by server are handled nicely

Application Services

News Content

- Changed provider to Guardian Service (instead of Current API)
- News Controller controls the different routes

AI & Embeddings

- Google generative AI service (Gemini) for summaries, AI explanations
- Stores embeddings in the Database

User & Profile

- Auth, admin and user management controllers ensure right access to user profiles on database
- Profile service changes individual profiles as users sign up

Agile Alligators
Feb 11, 2026

External Services

- Guardian news API service – API key used
- Google generative AI service – API key + tokens used

Data Layer

- PostgreSQL / Yugabyte database
- Migration folder used to control/edit schema through the code